COMPACTIFICATIONS GLUE OVER AN ELEMENTARY DISTINGUISHED SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3492-3500 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 3502-3755, the complete author proof (252 body lines).

Editorial note. Navigation stubs: 16 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_spaces-pushouts.json (pb=11519, stmt 3492-3500, proof 3502-3755).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-two-compactifications' also exists in stacks-flat, stacks-snapshot-20261011 — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0F48: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3340–3356. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F47: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3297–3305. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F45: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3219–3224. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F4A: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3459–3467. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0DVJ: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1764–1778. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F49: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3426–3435. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F46: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3240–3252. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ASK: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3755–3755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 088Q: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3755–3755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03KN: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3755–3755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03KA: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3755–3755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 088G: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3755–3755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 089G: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3755–3755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03KU: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3755–3755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NX: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3755–3755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A3W: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3755–3755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let B be an algebraic space of finite type over $\mathbf{Z}$. Let U be an algebraic space of finite type and separated over B. Let $(U_2 \subset U, f : U_1 \to U)$ be an elementary distinguished square. Assume U_1 and U_2 have compactifications over B and $U_1 \times_U U_2 \to U$ has dense image. Then U has a compactification over B.*

Proof. Choose a compactification $U_i \subset X_i$ over B for $i = 1, 2$. We may assume U_i is scheme theoretically dense in X_i. We may assume there is an open $V_i \subset X_i$ and a proper morphism $\psi_i : V_i \to U$ extending $U_i \to U$, see Lemma 0F4A. Picture

$$\begin{array}{ccccc} U_i & \longrightarrow & V_i & \longrightarrow & X_i \\ \downarrow & \swarrow_{\psi_i} & & & \\ U & & & & \end{array}$$

Denote $Z_1 \subset U$ the reduced closed subspace corresponding to the closed subset $|U| \setminus |U_2|$. Recall that $f^{-1}Z_1$ is a closed subspace of U_1 mapping isomorphically to Z_1. Denote $Z_2 \subset U$ the reduced closed subspace corresponding to the closed subset $|U| \setminus \mathrm{Im}(|f|) = |U_2| \setminus \mathrm{Im}(|U_1 \times_U U_2| \to |U_2|)$. Thus we have

$$U = U_2 \amalg Z_1 = Z_2 \amalg \mathrm{Im}(f) = Z_2 \amalg \mathrm{Im}(U_1 \times_U U_2 \to U_2) \amalg Z_1$$

set theoretically. Denote $Z_{i,i} \subset V_i$ the inverse image of Z_i under ψ_i. Observe that ψ_2 is an isomorphism over an open neighbourhood of Z_2. Observe that $Z_{1,1} = \psi_1^{-1}Z_1 = f^{-1}Z_1 \amalg T$ for some closed subspace $T \subset V_1$ disjoint from $f^{-1}Z_1$ and furthermore ψ_1 is étale along $f^{-1}Z_1$. Denote $Z_{i,j} \subset V_i$ the inverse image of Z_j under ψ_i. Observe that $\psi_i : Z_{i,j} \to Z_j$ is a proper morphism. Since Z_i and Z_j are disjoint closed subspaces of U, we see that $Z_{i,i}$ and $Z_{i,j}$ are disjoint closed subspaces of V_i.

Denote $\overline{Z}_{i,i}$ and $\overline{Z}_{i,j}$ the scheme theoretic images of $Z_{i,i}$ and $Z_{i,j}$ in X_i. We recall that $|Z_{i,j}|$ is dense in $|\overline{Z}_{i,j}|$, see Morphisms of Spaces, Lemma 088G. After replacing X_i by a V_i-admissible blowup we may assume that $\overline{Z}_{i,i}$ and $\overline{Z}_{i,j}$ are disjoint, see Lemma 0F46. We assume this holds for both X_1 and X_2. Observe that this property is preserved if we replace X_i by a further V_i-admissible blowup. Hence we may replace X_1 by another V_1-admissible blowup and assume $|\overline{Z}_{1,1}|$ is the disjoint union of the closures of $|T|$ and $|f^{-1}Z_1|$ in $|X_1|$.

Set $V_{12} = V_1 \times_U V_2$. We have an immersion $V_{12} \to X_1 \times_B X_2$ which is the composition of the closed immersion $V_{12} = V_1 \times_U V_2 \to V_1 \times_B V_2$ (Morphisms of Spaces, Lemma 03KN) and the open immersion $V_1 \times_B V_2 \to X_1 \times_B X_2$. Let $X_{12} \subset X_1 \times_B X_2$ be the scheme theoretic image of $V_{12} \to X_1 \times_B X_2$. The projection morphisms

$$p_1 : X_{12} \to X_1 \quad \text{and} \quad p_2 : X_{12} \to X_2$$

are proper as X_1 and X_2 are proper over B. If we replace X_1 by a V_1-admissible blowing up, then X_{12} is replaced by the strict transform with respect to this blowing up, see Lemma 0F49.

Denote $\psi : V_{12} \to U$ the compositions $\psi = \psi_1 \circ p_1|_{V_{12}} = \psi_2 \circ p_2|_{V_{12}}$. Consider the closed subspace

$$Z_{12,2} = (p_1|_{V_{12}})^{-1}Z_{1,2} = (p_2|_{V_{12}})^{-1}Z_{2,2} = \psi^{-1}Z_2 \subset V_{12}$$

The morphism $p_1|_{V_{12}} : V_{12} \to V_1$ is an isomorphism over an open neighbourhood of $Z_{1,2}$ because $\psi_2 : V_2 \to U$ is an isomorphism over an open neighbourhood of Z_2 and $V_{12} = V_1 \times_U V_2$. By Lemma 0F47 there exists a V_1-admissible blowing up $X_1' \to X_1$ such that the strict transform $p_1' : X_{12}' \to X_1'$ of p_1 is an isomorphism over an open neighbourhood of the closure of $|Z_{1,2}|$ in $|X_1'|$. After replacing X_1 by X_1' and X_{12} by X_{12}' we may assume that p_1 is an isomorphism over an open neighbourhood of $|\overline{Z}_{1,2}|$.

The result of the previous paragraph tells us that

$$X_{12} \cap (\overline{Z}_{1,2} \times_B \overline{Z}_{2,1}) = \emptyset$$

where the intersection taken in $X_1 \times_B X_2$. Namely, the inverse image $p_1^{-1}\overline{Z}_{1,2}$ in X_{12} maps isomorphically to $\overline{Z}_{1,2}$. In particular, we see that $|Z_{12,2}|$ is dense in $|p_1^{-1}\overline{Z}_{1,2}|$. Thus p_2 maps $|p_1^{-1}\overline{Z}_{1,2}|$ into $|\overline{Z}_{2,2}|$. Since $|\overline{Z}_{2,2}| \cap |\overline{Z}_{2,1}| = \emptyset$ we conclude.

It turns out that we need to do one additional blowing up before we can conclude the argument. Namely, let $V_2 \subset W_2 \subset X_2$ be the open subspace with underlying topological space

$$|W_2| = |V_2| \cup (|X_2| \setminus |\overline{Z}_{2,1}|) = |X_2| \setminus \left(|\overline{Z}_{2,1}| \setminus |Z_{2,1}|\right)$$

Since $p_2(p_1^{-1}\overline{Z}_{1,2})$ is contained in W_2 (see above) we see that replacing X_2 by a W_2-admissible blowup and X_{21} by the corresponding strict transform will preserve the property of p_1 being an isomorphism over an open neighbourhood of $\overline{Z}_{1,2}$. Since $\overline{Z}_{2,1} \cap W_2 = \overline{Z}_{2,1} \cap V_2 = Z_{2,1}$ we see that $Z_{2,1}$ is a closed subspace of W_2 and V_2. Observe that $V_{12} = V_1 \times_U V_2 = p_1^{-1}(V_1) = p_2^{-1}(V_2)$ as open subspaces of X_{12} as it is the largest open subspace of X_{12} over which the morphism $\psi : V_{12} \to U$ extends; details omitted[1]. We have the following equalities of closed subspaces of V_{12}:

$$p_2^{-1}Z_{2,1} = p_2^{-1}\psi_2^{-1}Z_1 = p_1^{-1}\psi_1^{-1}Z_1 = p_1^{-1}Z_{1,1} = p_1^{-1}f^{-1}Z_1 \amalg p_1^{-1}T$$

Here and below we use the slight abuse of notation of writing p_2 in stead of the restriction of p_2 to V_{12}, etc. Since $p_2^{-1}(Z_{2,1})$ is a closed subspace of $p_2^{-1}(W_2)$ as $Z_{2,1}$ is a closed subspace of W_2 we conclude that also $p_1^{-1}f^{-1}Z_1$ is a closed subspace of $p_2^{-1}(W_2)$. Finally, the morphism $p_2 : X_{12} \to X_2$ is étale at points of $p_1^{-1}f^{-1}Z_1$ as ψ_1 is étale along $f^{-1}Z_1$ and $V_{12} = V_1 \times_U V_2$. Thus we may apply Lemma 0F48 to the morphism $p_2 : X_{12} \to X_2$, the open W_2, the closed subspace $Z_{2,1} \subset W_2$, and the closed subspace $p_1^{-1}f^{-1}Z_1 \subset p_2^{-1}(W_2)$. Hence after replacing X_2 by a W_2-admissible blowup and X_{12} by the corresponding strict transform, we obtain for every geometric point $\overline{y}$ of the closure of $|p_1^{-1}f^{-1}Z_1|$ a local ring map $\mathcal{O}_{X_{12},\overline{y}} \to \mathcal{O}$ such that $\mathcal{O}_{X_2,p_2(\overline{y})} \to \mathcal{O}$ is finite flat.

Consider the algebraic space

$$W_2 = U \coprod\nolimits_{U_2} (X_2 \setminus \overline{Z}_{2,1}),$$

and with $T \subset V_1$ as in the first paragraph the algebraic space

$$W_1 = U \coprod\nolimits_{U_1} (X_1 \setminus \overline{Z}_{1,2} \cup \overline{T}),$$

obtained by pushout, see Lemma 0DVJ. Let us apply Lemma 0F45 to see that $W_i \to B$ is separated. First, $U \to B$ and $X_i \to B$ are separated. Let us check the quasi-compact immersion $U_i \to U \times_B (X_i \setminus \overline{Z}_{i,j})$ is closed using the valuative criterion, see Morphisms of Spaces, Lemma 03KA. Choose a valuation ring A over B with fraction field K and compatible morphisms $(u, x_i) : \mathrm{Spec}(A) \to U \times_B X_i$ and $u_i : \mathrm{Spec}(K) \to U_i$. Since ψ_i is proper, we can find a unique $v_i : \mathrm{Spec}(A) \to V_i$ compatible with u and u_i. Since X_i is proper over B we see that $x_i = v_i$. If v_i does not factor through $U_i \subset V_i$, then we conclude that x_i maps the closed point of $\mathrm{Spec}(A)$ into $Z_{i,j}$ or T when $i = 1$. This finishes the proof because we removed $\overline{Z}_{i,j}$ and $\overline{T}$ in the construction of W_i.

[1]Namely, $V_1 \times_U V_2$ is proper over U so if ψ extends to a larger open of X_{12}, then $V_1 \times_U V_2$ would be closed in this open by Morphisms of Spaces, Lemma 04NX. Then we get equality as $V_{12} \subset X_{12}$ is dense.

On the other hand, for any valuation ring A over B with fraction field K and any morphism

$$\gamma : \mathrm{Spec}(K) \to \mathrm{Im}(U_1 \times_U U_2 \to U)$$

over B, we claim that after replacing A by an extension of valuation rings, there is an i and an extension of γ to a morphism $h_i : \mathrm{Spec}(A) \to W_i$. Namely, we first extend γ to a morphism $g_2 : \mathrm{Spec}(A) \to X_2$ using the valuative criterion of properness. If the image of g_2 does not meet $\overline{Z}_{2,1}$, then we obtain our morphism into W_2. Otherwise, denote $\overline{z} \in \overline{Z}_{2,1}$ a geometric point lying over the image of the closed point under g_2. We may lift this to a geometric point $\overline{y}$ of X_{12} in the closure of $|p_1^{-1}f^{-1}Z_1|$ because the map of spaces $|p_1^{-1}f^{-1}Z_1| \to |\overline{Z}_{2,1}|$ is closed with image containing the dense open $|Z_{2,1}|$. After replacing A by its strict henselization (More on Algebra, Lemma 0ASK) we get the following diagram

$$\begin{array}{ccccc} A & & \cdots\cdots\cdots\cdots\cdots\cdots & & A' \\ \uparrow & & & & \uparrow \\ \mathcal{O}_{X_2,\overline{z}} & \longrightarrow & \mathcal{O}_{X_{12},\overline{y}} & \longrightarrow & \mathcal{O} \end{array}$$

where $\mathcal{O}_{X_{12},\overline{y}} \to \mathcal{O}$ is the map we found in the 5th paragraph of the proof. Since the horizontal composition is finite and flat we can find an extension of valuation rings A'/A and dotted arrow making the diagram commute. After replacing A by A' this means that we obtain a lift $g_{12} : \mathrm{Spec}(A) \to X_{12}$ whose closed point maps into the closure of $|p_1^{-1}f^{-1}Z_1|$. Then $g_1 = p_1 \circ g_{12} : \mathrm{Spec}(A) \to X_1$ is a morphism whose closed point maps into the closure of $|f^{-1}Z_1|$. Since the closure of $|f^{-1}Z_1|$ is disjoint from the closure of $|T|$ and contained in $|\overline{Z}_{1,1}|$ which is disjoint from $|\overline{Z}_{1,2}|$ we conclude that g_1 defines a morphism $h_1 : \mathrm{Spec}(A) \to W_1$ as desired.

Consider a diagram

$$\begin{array}{ccccc} W_1' & \longrightarrow & W & \longleftarrow & W_2' \\ \downarrow & \nwarrow & \uparrow & \nearrow & \downarrow \\ W_1 & \longleftarrow & U & \longrightarrow & W_2 \end{array}$$

as in More on Morphisms of Spaces, Lemma 088Q. By the previous paragraph for every solid diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{\gamma} & W \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & B \end{array}$$

where $\mathrm{Im}(\gamma) \subset \mathrm{Im}(U_1 \times_U U_2 \to U)$ there is an i and an extension $h_i : \mathrm{Spec}(A) \to W_i$ of γ after possibly replacing A by an extension of valuation rings. Using the valuative criterion of properness for $W_i' \to W_i$, we can then lift h_i to $h_i' : \mathrm{Spec}(A) \to W_i'$. Hence the dotted arrow in the diagram exists after possibly extending A. Since W is separated over B, we see that the choice of extension isn't needed and the arrow is unique as well, see Morphisms of Spaces, Lemmas 0A3W and 03KU. Then finally the existence of the dotted arrow implies that $W \to B$ is universally closed by Morphisms of Spaces, Lemma 089G. As $W \to B$ is already of finite type and separated, we win. □